Supervised Learning Models: Preprocessing Cheat Sheet

Every model named in the module, grouped by topic.

Module topic	Model	Task	sklearn family	Class name	Scaling	Encoding	Missing values
2. Regression	Linear Regression	Regression	linear_model	LinearRegression	Helpful	Yes	Must fill
2. Regression	Multiple Linear Regression	Regression	linear_model	LinearRegression (many features)	Helpful	Yes	Must fill
3. Classification	Logistic Regression	Classification	linear_model	LogisticRegression	Yes	Yes	Must fill
3. Classification	KNN	Classification	neighbors	KNeighborsClassifier	**Must**	Yes	Must fill
3. Classification	Naive Bayes	Classification	naive_bayes	GaussianNB	No	Yes	Must fill
4. Tree-based	Decision Tree	Both	tree	DecisionTreeClassifier / DecisionTreeRegressor	No	Yes	Must fill
4. Tree-based	Random Forest	Both	ensemble	RandomForestClassifier / RandomForestRegressor	No	Yes	Must fill
5. Boosting	XGBoost	Both	xgboost (separate library)	XGBClassifier / XGBRegressor	No	Yes	Can handle itself

Easy rules

- **Linear, Logistic, KNN:** scale the numeric columns.
- **Decision Tree, Random Forest, XGBoost, Naive Bayes:** no scaling needed.
- **Every model** needs text columns converted to numbers (encoding).
- **Only XGBoost** handles missing values by itself. For all others, fill them with SimpleImputer.

Quick notes

- Linear and Multiple Linear Regression use the same class. "Multiple" just means more than one input column.
- LogisticRegression is a classification model, even though its name says "regression".
- Topics 1, 6 and 7 (fundamentals, workflows, feature importance) have no models of their own. They apply to all models above.